

Solution Code - Spring Core and Maven - Bean_Lifecycle

Below is the complete functional source code implementation for this assignment. All logic, validation rules, and structural requirements have been satisfied.

Source File: src\main\java\com\library\BookService.java

```
1 | package com.library;
2 |
3 | public class BookService {
4 |     public void init() {
5 |         System.out.println("Bean Lifecycle Init: Setting up resources.");
6 |     }
7 |     public void destroy() {
8 |         System.out.println("Bean Lifecycle Destroy: Releasing connection channels.");
9 |     }
10 | }
```

Source File: src\main\java\com\library\LibraryApp.java

```
1 | package com.library;
2 |
3 | import org.springframework.context.support.ClassPathXmlApplicationContext;
4 |
5 | public class LibraryApp {
6 |     public static void main(String[] args) {
7 |         |
8 |         ClassPathXmlApplicationContext context = new
ClassPathXmlApplicationContext("applicationContext.xml");
9 |         BookService service = (BookService) context.getBean("bookService");
10 |         context.close(); // Triggers destroy methods
11 |     }
```

Source File: src\main\resources\applicationContext.xml

```
1 | <beans xmlns="http://www.springframework.org/schema/beans"
2 |       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
3 |       xsi:schemaLocation="http://www.springframework.org/schema/beans
4 |       http://www.springframework.org/schema/beans/spring-beans.xsd">
5 |
6 |     <bean id="bookService" class="com.library.BookService" init-method="init" destroy-method="destroy" />
7 | </beans>
```